Static Timing Analysis

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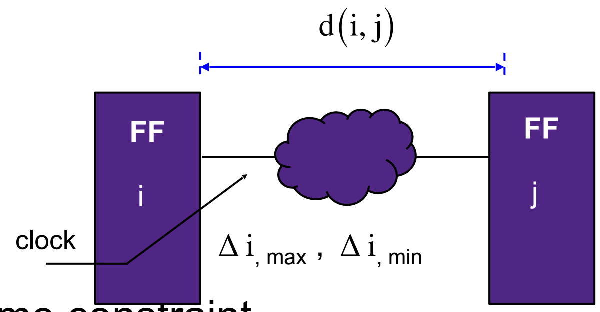
4. Delay Calculation

Problems in Edge-triggered Circuits



- Setup time constraint

  $\Delta i_{,max} + dmax(i,j) \leq P - T_{S_j}$

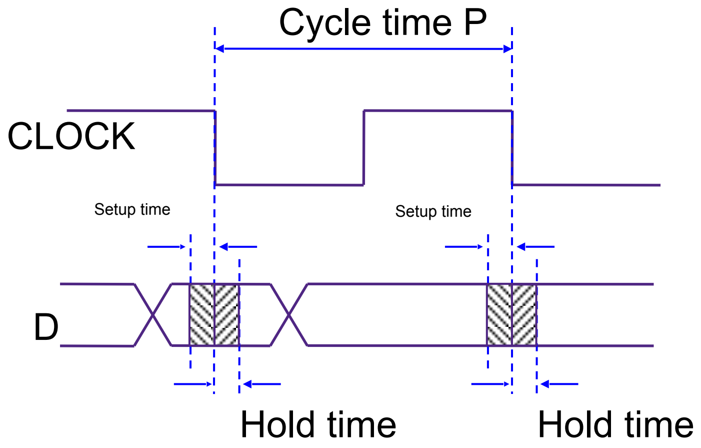
  i.e., $dmax(i,j) \leq P - T_{S_j} - \Delta i_{,max}$
- Long path constraint
- Hold time constraint

  $\Delta i_{,min} + dmin(i,j) \geq T_{h_j}$

  i.e., $dmin(i,j) \geq T_{h_j} - \Delta i_{,min}$
- Short path

$dmax(i,j)$ - largest delay
$dmin(i,j)$ - smallest delay
$dmin(i,j) \leq d(i,j) \leq dmax(i,j)$

T_{S_j} - setup time
T_{h_j} - hold time
$\Delta i_{,max}$ – maximum delay
$\Delta i_{,min}$ – minimum delay

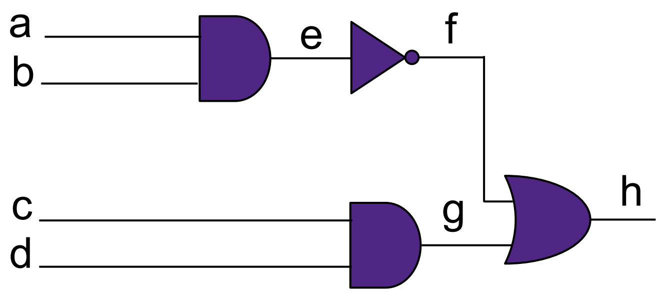


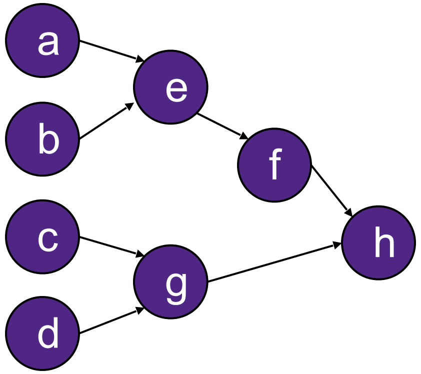
Required and Arrival Times

- Arrival time constraints at each output node
- It is useful to know the required time at each node
  - If the arrival time exceeds the required time, the path must be sped up

Path-based Timing Calculation

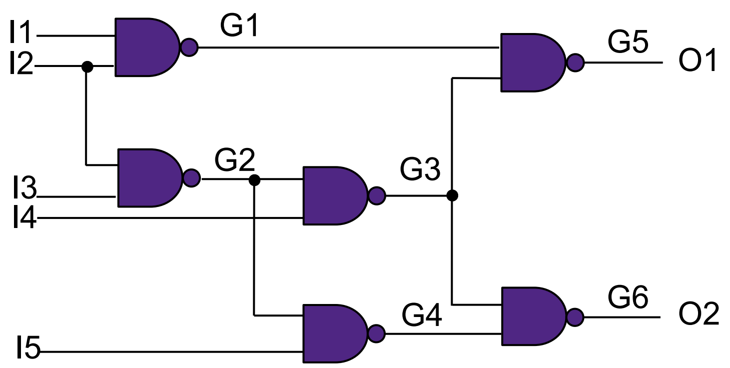
- Calculates minimum and maximum path delay costs
- The timing analyzer represents a netlist as a directed graph
  - Nodes in the graph
  - Edges represent
    - Net delay – interconnect delay between a driver pin and a load pin its fanout
    - Cell delay – timing delay between an input pin and an output pin of a cell

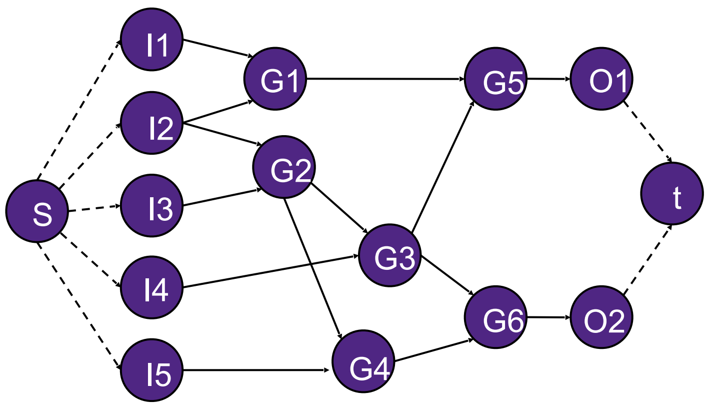




Representation of Circuits

- Combinational logic circuit may be represented as a timing graph G = (V,E)
  - V, the vertex set, are the logic gates in the circuit and the primary inputs and outputs of the circuit
- Vertices, u and v $\in$ G , are connected by a directed edge e(u, v) $\in$ E
  - Connection from the output of the element represented by vertex u to the input of the element represented by vertex v
- Circuit is represented by directed acyclic graph (DAG)
  - Do not have any cycles





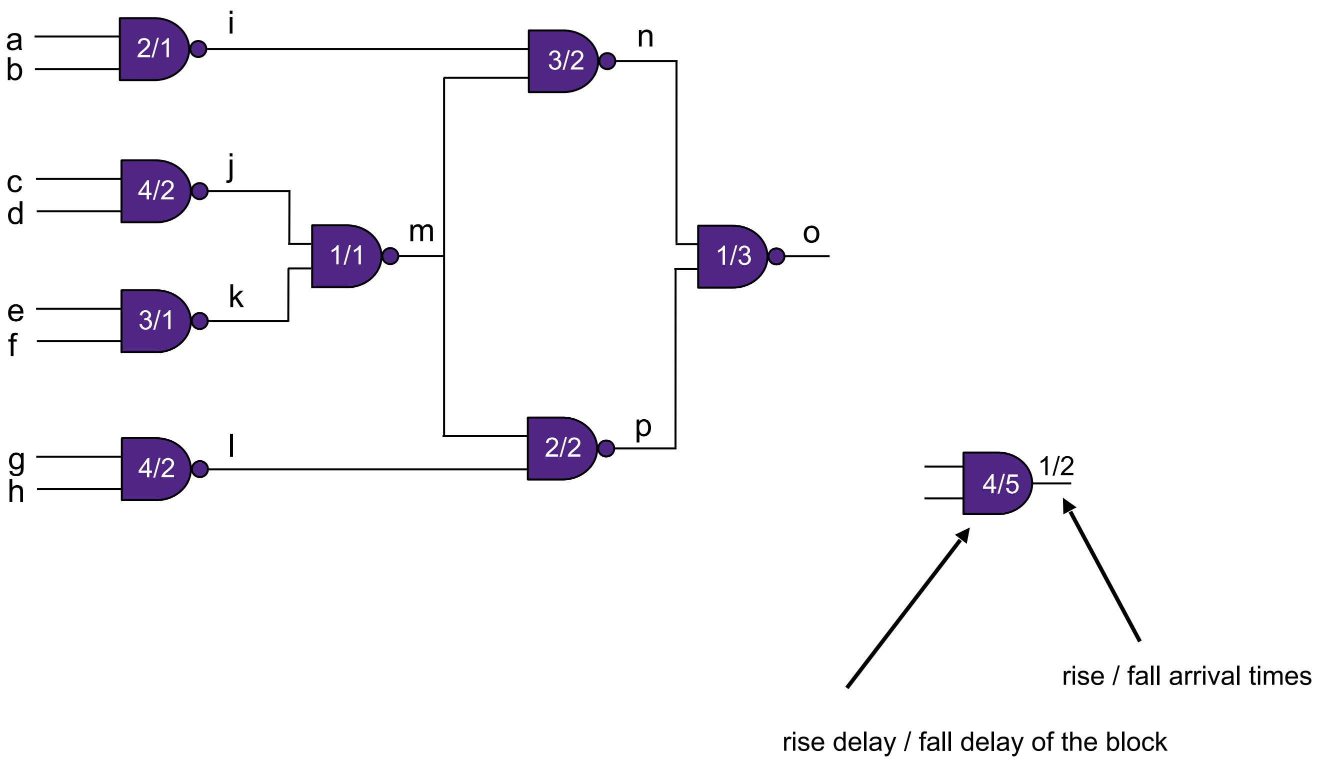
Critical Path Method

- CPM (Critical Path Method) for Timing graph G = (V,E)

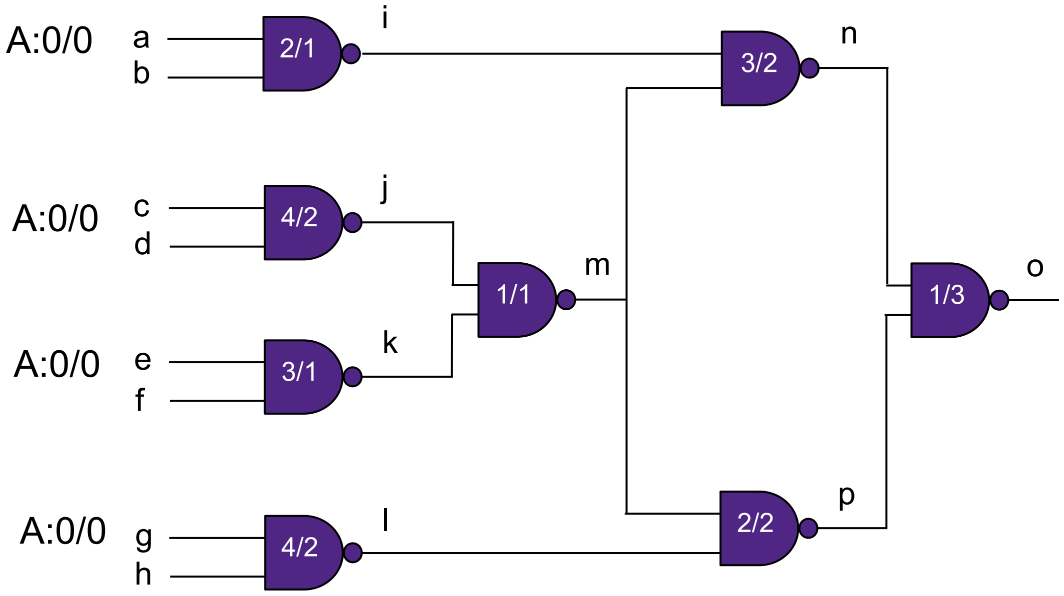
Computing the worst-case rise and fall arrival times at each intermediate node, and output(s) of a circuit

```
Q = ∅ ;
for all vertices i ∈ V
  n_visited_inputs [i] = 0 ;
        /* Add a vertex to the tail of Q if all inputs are ready */
        for all primary inputs i
            /* Fanout gates of i */
            for all vertices j such that (i → j) ∈ E
             if(++n_visited_inputs[j] == n_inputs[j]) addQ(j,Q) ;
while (Q ≠ ∅) {
     g = top (Q) ;
     remove (g,Q) ;
     compute_delay[g]
     /* Fanout gates of g */
     for all vertices k such that (g → k) ∈ E
           if(++n_visited_inputs[k] == n_inputs[k]) addQ(k,Q) ;
}
```

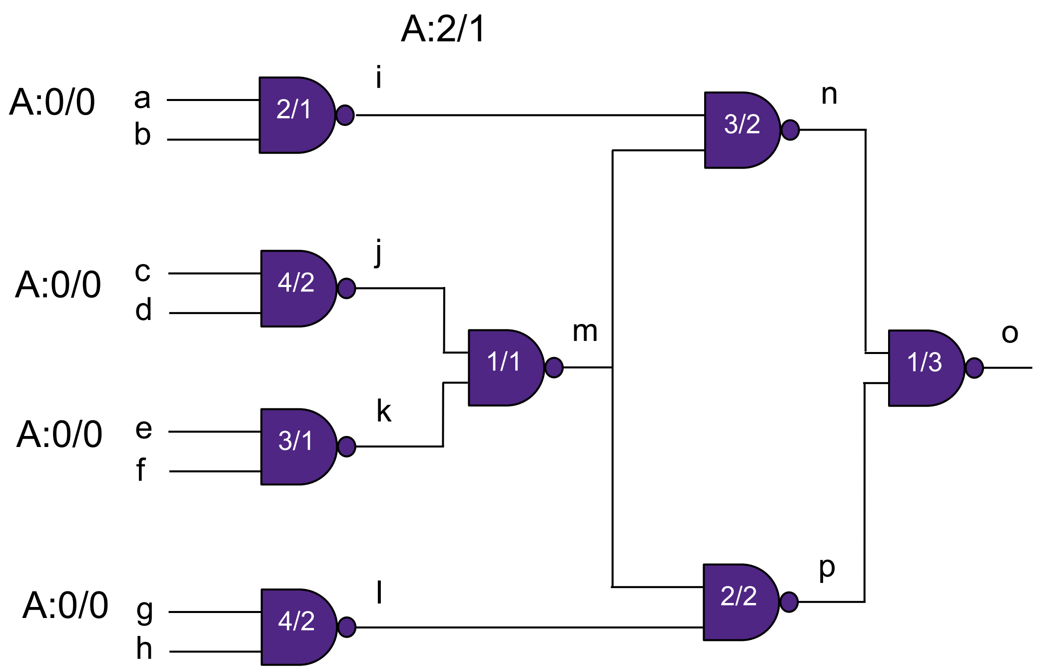
Given Circuit with Delays of Components



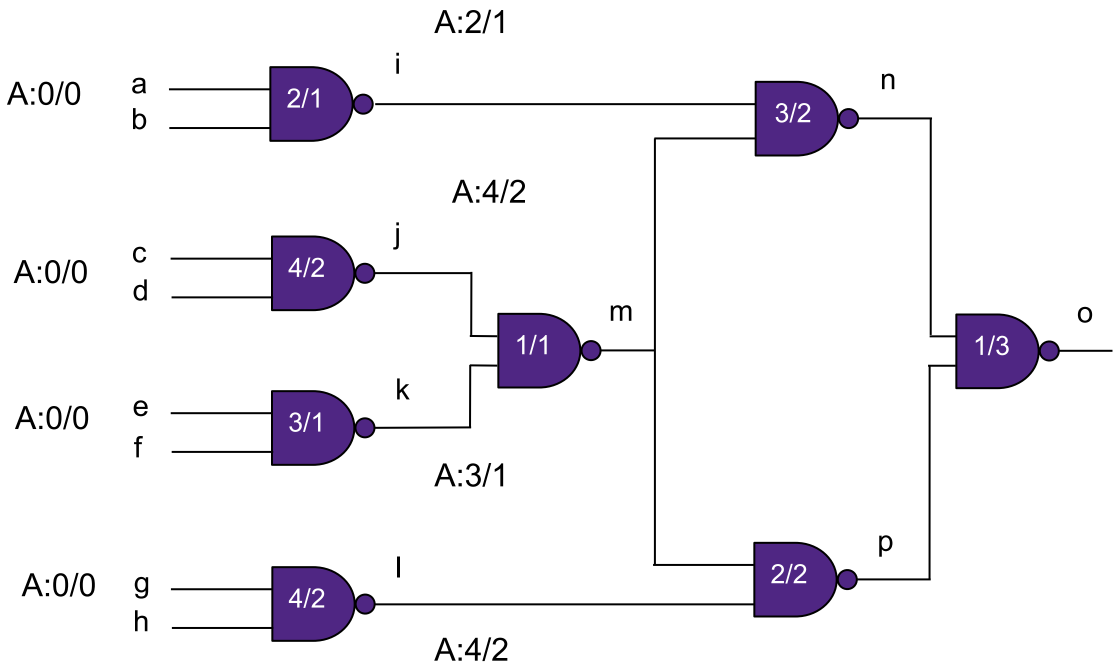
CPM Arrival Time Calculation Steps



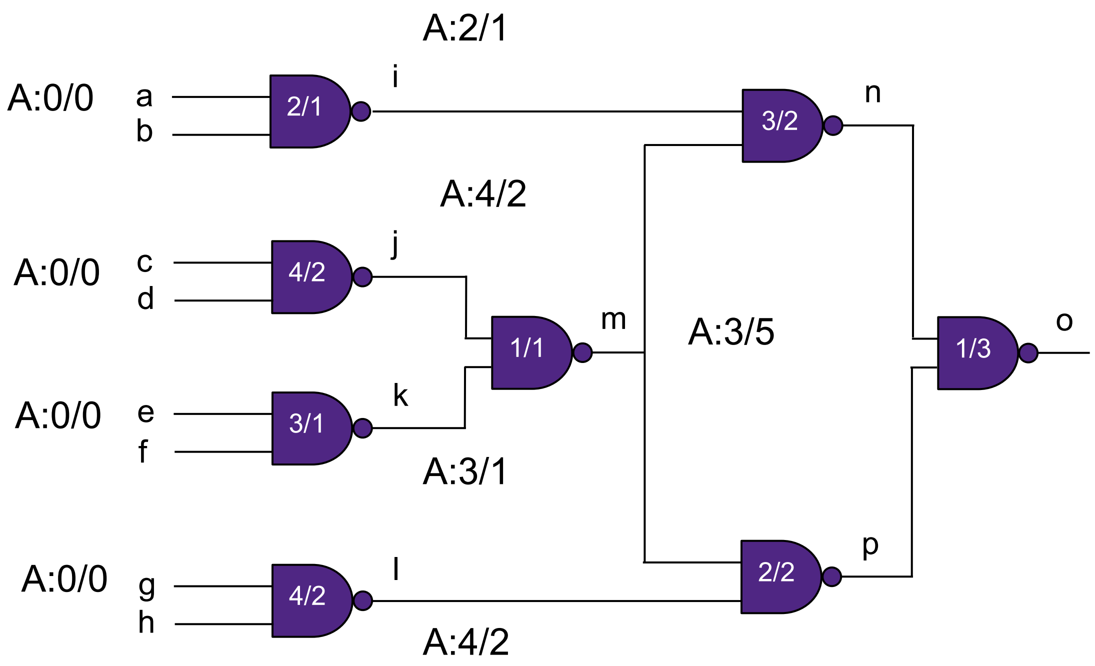
CPM Arrival Time Calculation Steps (2)



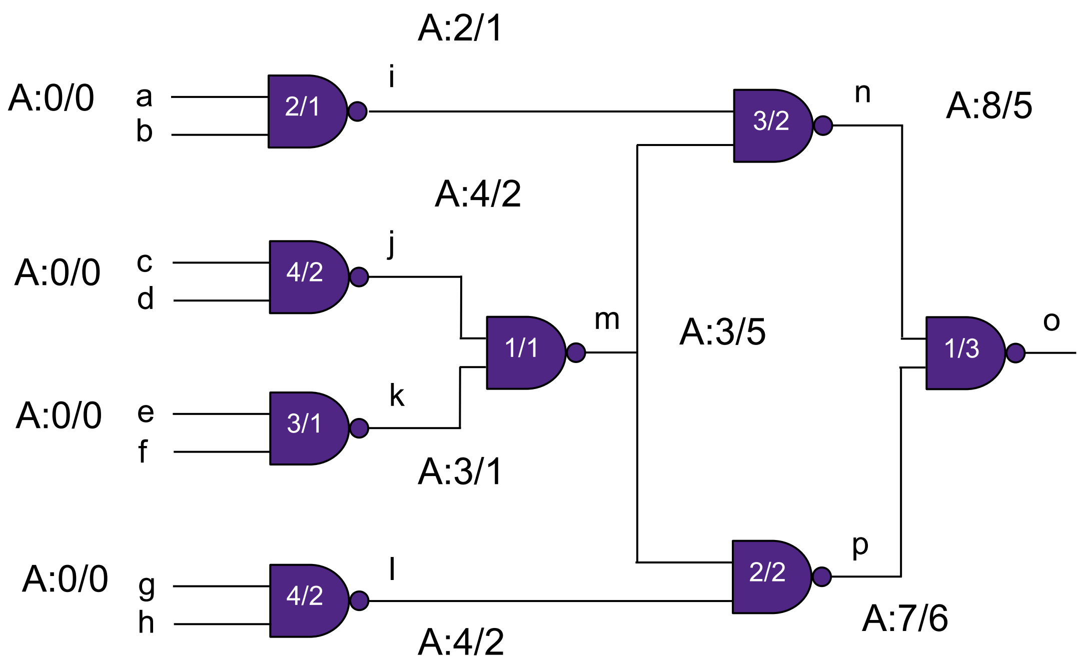
CPM Arrival Time Calculation Steps (3)



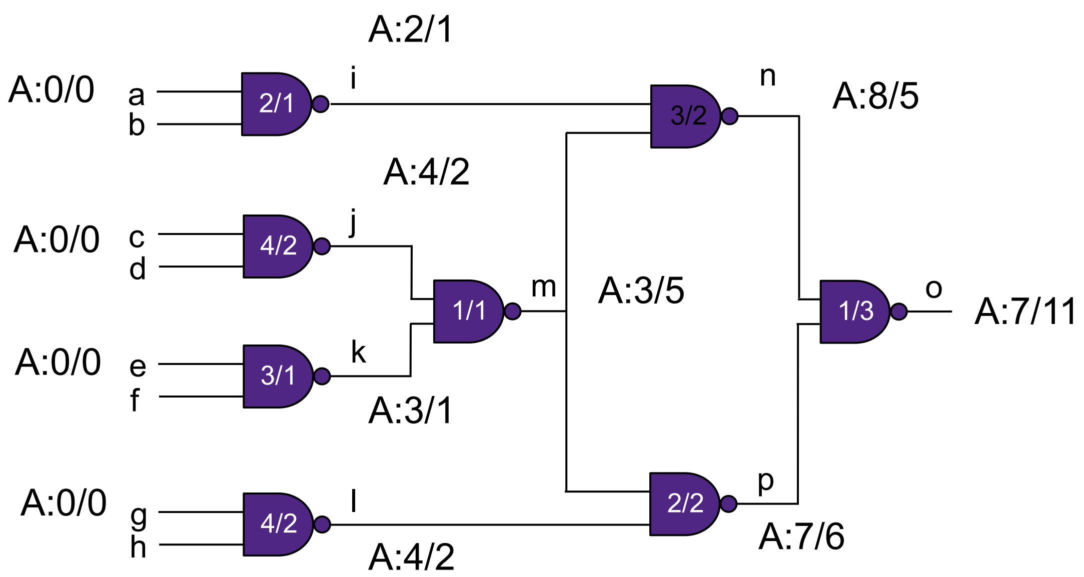
CPM Arrival Time Calculation Steps (4)



CPM Arrival Time Calculation Steps (5)

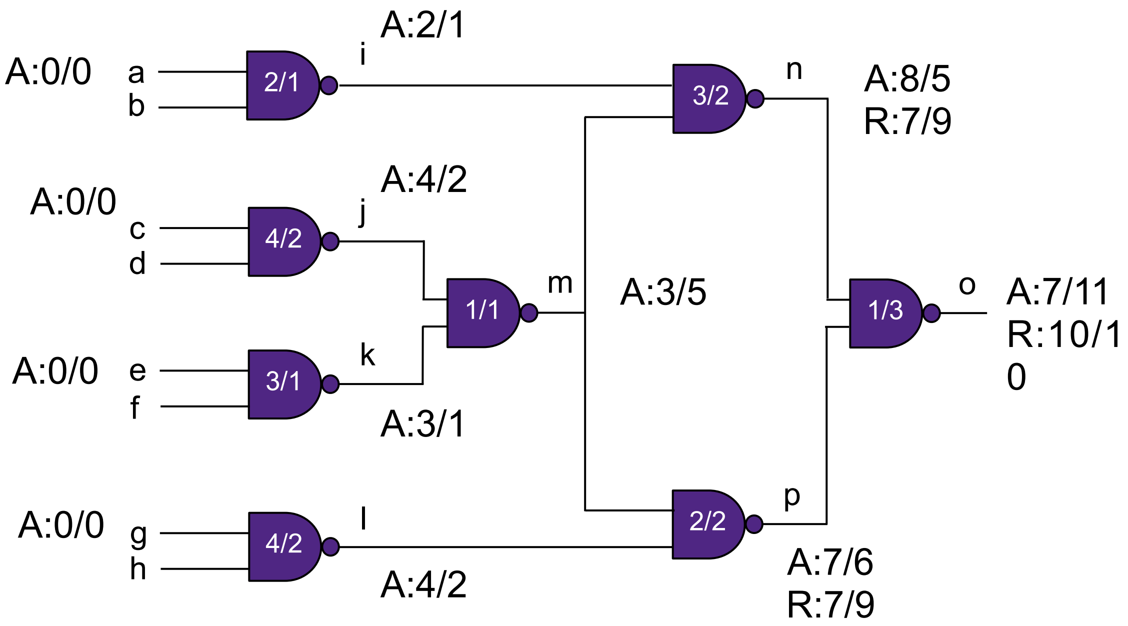


CPM Arrival Time Calculation Steps (6)



Required Time Calculation Steps

- At each primary output, the rise/fall required times are set, then backward topological traversal is carried out.



Required Time Calculation Steps (2)

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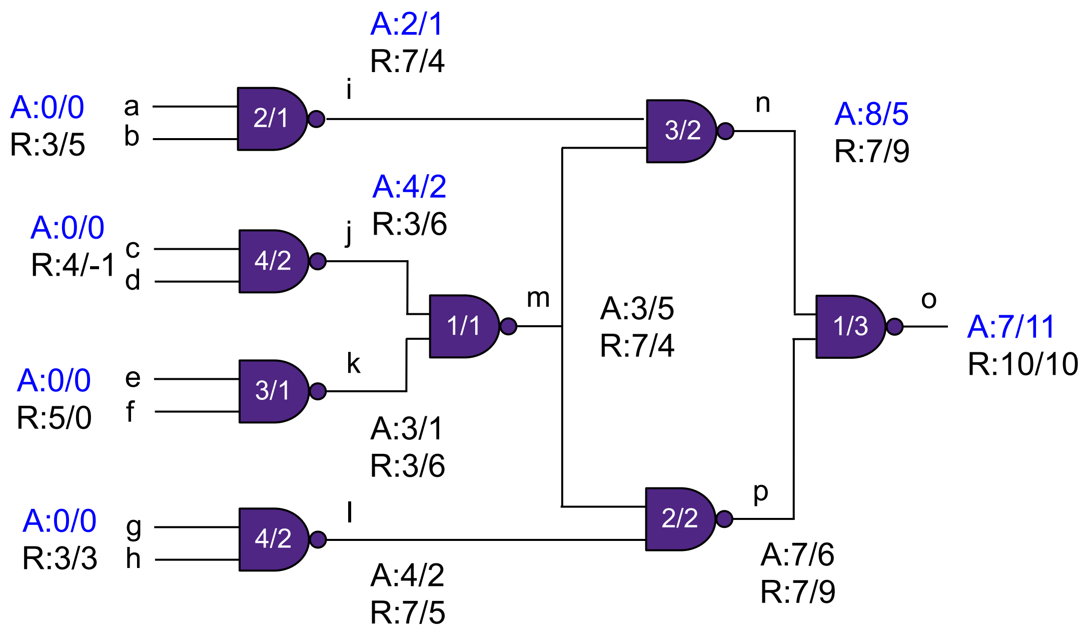

Slack Calculation

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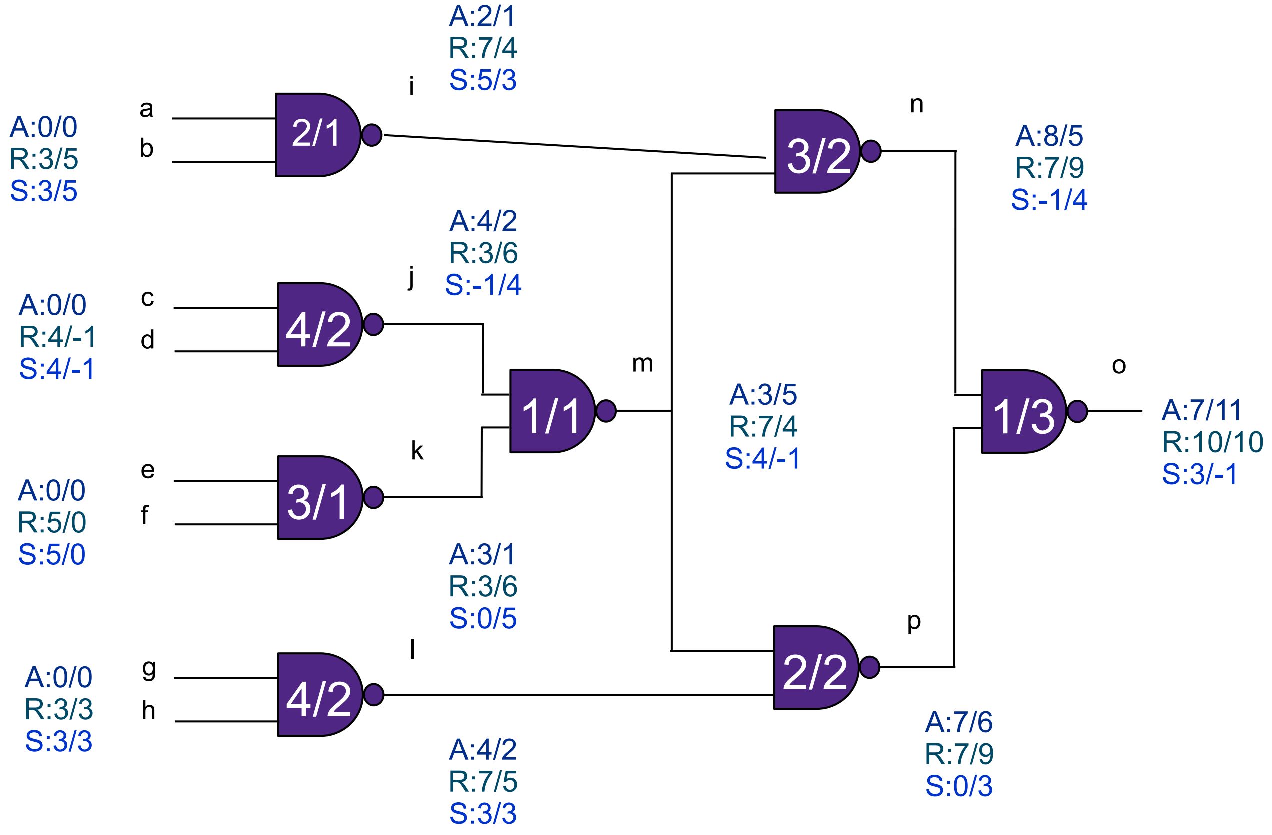

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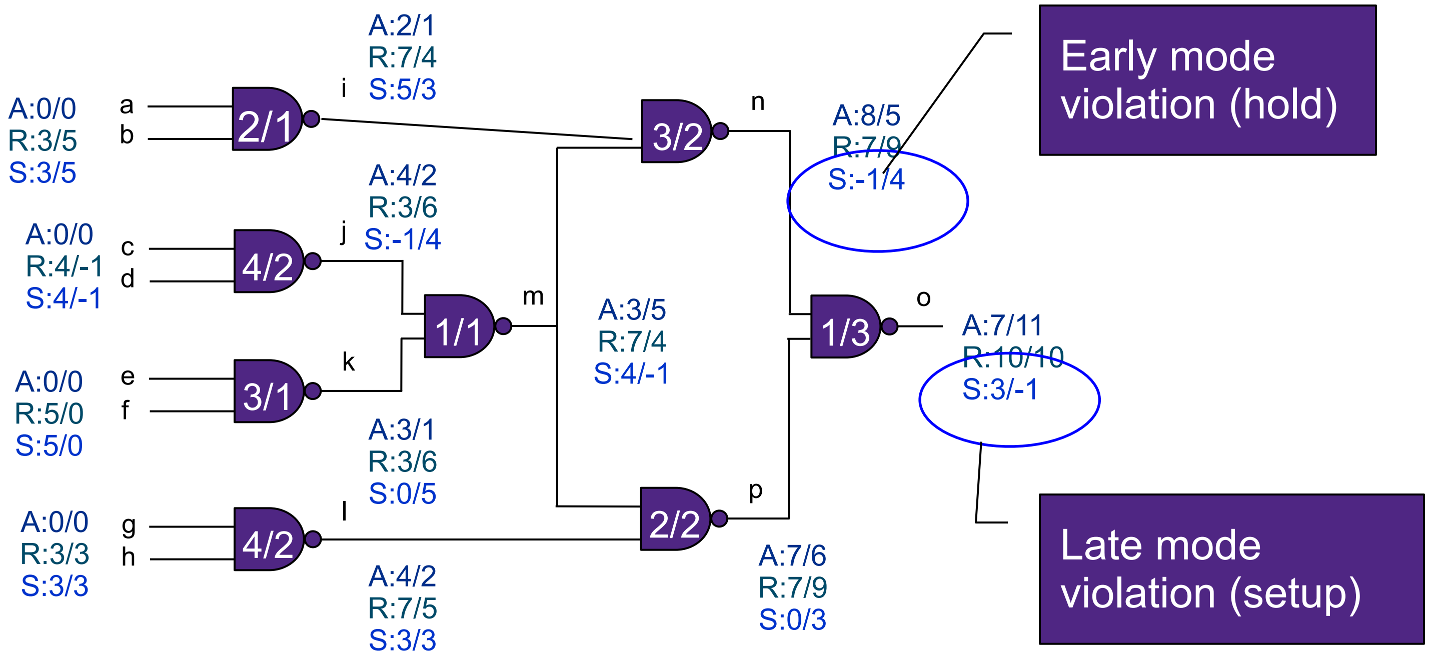
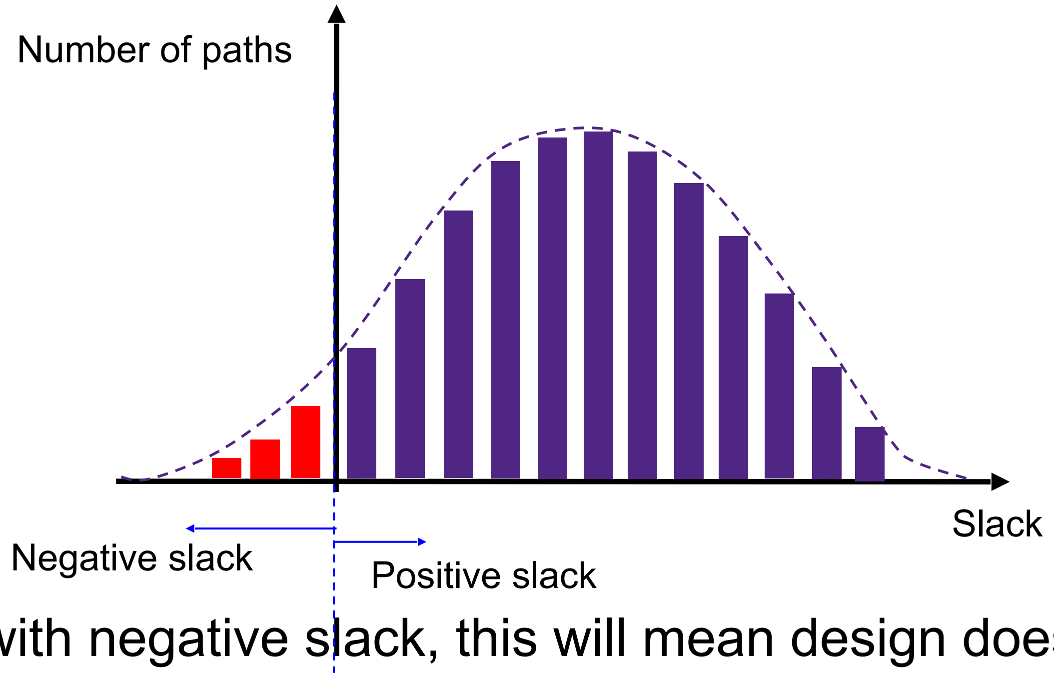


Path Slack Histogram



- If there is no path with negative slack, this will mean design does not have timing violations.

Thank You